Operating Instructions

ASSA ABLOY

MCS Control System



6410R0331

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These operating instructions are an integral part of the door. The original version of these instructions is in German.

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ASSA ABLOY

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1 Introduction

These operating instructions apply to the use of relay control system MCS-R and frequency converter control system MCS-FC with high-speed doors. They are part of the product and describe its intended and safe use throughout the whole product life cycle.

Before the installation of the MCS, read the installation and operating instructions carefully and familiarize yourself with their contents.

The manufacturer is not liable for any damage caused by failure to observe the operating and safety instructions or by unintended use of the MCS.

1.1 Applicable documents

These operating instructions are only valid in combination with the operating instructions for Albany high-speed doors from ASSA ABLOY Entrance Systems and the inspection book. Both documents are supplied with the product.

1.2 Declaration of conformity

The control unit is an integral part of the door. The declaration of conformity applies to the high-speed door as well as the control unit. It is included in the inspection book supplied with the door.

The control unit is designed in compliance with the requirements of the EN 13241-1 standard for door products and the subordinate standards mentioned in this standard.

Modifications that affect the technical data specified in the operating instructions or the intended use of the unit, i.e. modifications that alter it significantly, invalidate the manufacturer’s declaration of conformity.

1.3 Amendments and validity

The information contained in these instructions constitutes the technical specifications applicable at time of publication. Significant changes will be incorporated in a new edition of the operating instructions. The document and version number of these instructions can be found in the footer.

1.4 Document storage

Keep these operating instructions near the door.

1.5 Target group

These operating instructions are addressed to trained, qualified and authorised personnel. All setup configurations, and in particular repairs and maintenance, may only be performed by employees of the manufacturer or other trained specialists.

2 Safety instructions

Before installation, operation, service or maintenance work on this high-speed door or its electrical components, it is mandatory to read and to comply with the safety instructions.

In addition to the general safety instructions in this section, always observe the safety instructions relating to specific actions included in other sections. In particular, the general safety instructions in the operating instructions for the door are also applicable.

Non-observance of these safety instructions can lead to damage of the door, injury or death.

2.1 General safety instructions

- Observe the general, legal and other nationally valid regulations for accident prevention and the accepted technical rules for safety-oriented work.
- All work on electrical systems and connections may be performed only by qualified personnel with appropriate authorisation.
- The electrical work must be carried out according to the applicable local standards and legal regulations.
- Before performing any work on the control unit, cables, connectors or electrically live components, always disconnect and secure the CEE plug or turn of main switch.
- Do not operate the door unless its electrical supply is equipped with a main disconnect device (such as a CEE plug or main switch) that allows the electrical supply to the door to be disconnected. This device must be easily accessible and located between 0.6 and 1.7 metres above the access level.
- Make no change to the control unit or the safety devices which may affect the safety of the door.
- In order to avoid damage by electronic discharge, do not touch the circuit board of the control system, the display and the auxiliary modules without appropriate ESD protective equipment.
- The control system is protected by a freely selectable access code to prevent unauthorised access. Detailed information about the access code can be found in Section 3.2.2.

2.2 Safety instructions for operating the door

- The door is assembled in accordance with the installation instructions.
- All wiring between the door, the control unit and the drive is connected properly.
- All safety devices and protection equipment are installed as intended and are functional.
- Safety devices are not modified or non-operational.
- The user must be instructed to leave the mains power on at all times to avoid the build-up of condensation in the control system housing.

2.3 Safety instructions for installation and maintenance

- Ensure that neither persons nor vehicles pass through the door or are in the door hazard area during installation and maintenance.
- Dangerous contact voltages are present at the frequency converter up to 180 seconds after switching off/disconnecting the supply voltage (discharge of the capacitors). You must therefore wait at least 180 seconds after disconnecting the supply voltage before working on electrical parts, wiring or opening the frequency converter housing.
  Non-observance of this rule can lead to electric shock, serious injury or death.
- You should generally avoid touching electronic components on the MCS board or auxiliary modules.
- When opening the housing cover of the MCS or the frequency converter, ensure that the control system is protected against moisture.

2.4 Safety instructions for “hold to run mode”

- In hold to run mode, all safety devices are disabled. The door moves as long as the opening or closing key is pressed and it has not reached its upper or lower end position. The door does not stop until it reaches these positions.
- Operate the door only when it is visible.
- Operate the door only when no persons or objects are present in the door hazard area.

3 Product Description

The MCS control system is designed to be used with Albany high-speed doors from ASSA ABLOY Entrance Systems. The MCS satisfies the valid and applicable European CE requirements, as proved and confirmed by independent testing institutes. The control system contains all necessary components to operate these doors with due consideration of all safety criteria.

The MCS-R control system is an electronic relay controller. The MCS-FC consists of the MCS-R control unit extended by an external frequency converter to act as a frequency converter control system. The MCS-FC with the frequency converter enables soft acceleration and deceleration of the door. This protects the door's mechanism against hard impacts and increases its speed.

3.1 MCS main unit with control box



Figure 1: MCS-R control system



Figure 2: MCS-FC (MCS-R with external frequency converter)

The control system and the frequency converter are installed in separate plastic housings. Pre-cut openings can be broken out for the cable glands.

Inside the housing, the following components are located on the control board:

- The plug-in connection terminals for the inputs and outputs of the control system.
- The connection terminals for the power supply, the motor and brake cables.
- The power supply for the complete microprocessor and the external 12/24V loads.
- The dry contacts

3.1.1 Safety devices of the control system

The closing sequence of an automatic door is monitored by various safety devices in order to avoid accidents. The safety devices and functions incorporated in the control unit are described below.

The doors are also equipped with other safety devices, depending on the door model. See the 'Safety Devices' section of the RapidRoll user manual for information on the safety devices provided with the various models.

A pre-running safety photocell, a safety light curtain in the side frame or an electrical safety edge can be used as electric safety devices. They are complemented by a door line photocell in the side frame (except with the safety light curtain).

The electrical safety circuits of the MCS control system satisfy the requirements of the EN 13241-1 product standard for doors.

3.1.1.1 Testing the safety devices

Before each closing movement of the door, the fault free operation of the safety devices and the door line photocell are tested. If a fault is detected, no release of the closing movement occurs and the control unit sends an error message. In case of an error, the control unit switches to hold to run mode which allows manual operation of the door.



Observe the safety instructions for “hold to run mode” (Section 2.4).

3.1.1.2 Electrical safety edge (only for some door types)

The electrical safety edge acts as a closing edge protection. It consists of a rubber profile fitted to the main closing edge of the door. Any deformation of this profile initiates an electrical pulse which immediately stops the door while it is closing, reverses it and locks it at the upper end position. Locking is indicated during actuation by the following message on the display:

SAFE.EDGE 1
TRIG.

The mode then appears in the display on the top line, and a line is shown instead of the hold open time in the bottom line.

The door locking state is released by pressing the “Open” or “Close” keys.

However, this function can be adjusted so that the door closes automatically at a set time after the electrical safety edge has been activated (Section 4.8.17).

Just before the lower end position of the door (max. 50 mm/approx. 1.9”), the electrical safety edge is switched off because it would otherwise be activated via the ground. Details of the switch-off point of the contact edge may be found in section 4.8.3.

3.1.1.3 Door line photocell

The door line photocell is mounted in the lower area of the side frame of the door where it is protected. The correct operation is also monitored by the control system. Activation of the photocell while the door is open resets and starts the hold-open time

or the close-after-passage time. Activation of the photocell while the door is closing causes the door to stop immediately and then reverse until it reaches its upper end position. Unlike for the electrical safety edge, the door is not locked in this end position but closes again automatically after the hold-open or close-after-passage time if the door was closing in auto close mode while the photocell was activated. Otherwise it will stay open. As an option, a light curtain may be fitted on some door types instead of the door line photocell.

3.1.1.4 Emergency stop

The MCS control unit makes provision for connecting an emergency stop device corresponding to Stop Category 0 (stopping by immediate interruption of the power supply to the motor).

3.1.1.5 Emergency hand chain operator (KNH)

An optional emergency hand chain operator can be included in the safety chain at the digital limit switch in the motor (see Section 4.5).

3.1.2 Operating modes

The control unit has several operating modes which can be selected by means of menus. The main operating modes are described below.

3.1.2.1 AUTOMATIC UP/DOWN

After an Open pulse, the door moves until it reaches its set door-open end position. The closing sequence starts after the hold-open/close-after-passage time expires. The door closes until it reaches its lower end position. The hold-open and close-after-passage times can be set in the special setup menu (Section 4.8.16).

If the door is closed, an Open pulse from the external actuator or the Open button on the control system triggers the door to open immediately. If the door is open, the Open pulse restarts the hold-open time.

In response to a Stop command, either externally or on the control unit, the door stops immediately and remains where it is until motion in the desired direction is initiated by another actuator. The control panel stop can be acknowledged by an Open - or Close command on the control box or via external actuators.

If the door is in the fully opened state, a Stop signal terminates automatic closing and the door remains open until the next pulse. If this is followed by a Close pulse, the closing sequence starts immediately. An Open pulse restarts the hold-open time and triggers the door to close automatically after the time expires.

Interruption of the door line photocell or other safety devices causes the door to reverse to the open position and to close after the hold-open time.

3.1.2.2 Toggle circuit / manual closing

The toggle circuit / manual closing option allows the door to be opened and closed via a single actuator. As long as the door is not fully open, every incoming pulse is interpreted as an Open pulse. A pulse received when the door is open starts a closing sequence. For safety reasons, only manual actuators (pull switches or push

button) should be used with this function. The input KL20/7 is used for the toggle circuit.

Interruption of the door line photocell or other safety devices causes the door to reverse and remain in the open position. Closing of the door is started by a new pulse.

3.1.2.3 Close-after-passage function

If the close-after-passage function is activated, the door can be opened by an Open actuator. The closing sequence is initiated by interruption of the door line photocell and the door closes after the adjusted time (clearing-/close-after-passage-time). The door closes after the hold open time or after a close pulse if the door line photocell is not interrupted.

The settings are described in Section 4.8.16.

3.1.2.4 Hold to run mode

The door moves as long as the Open/Close button is pressed and the door has not reached its upper or lower end position. The door switches off automatically when it reaches these end positions. All safety devices are disabled in hold to run mode.



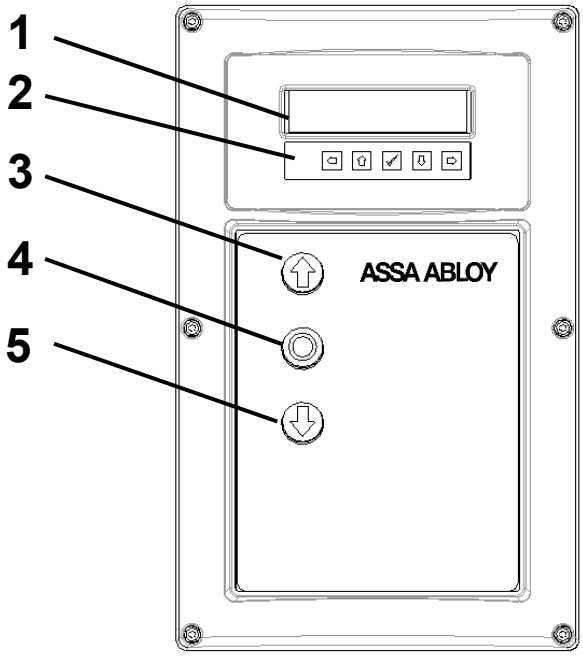
Observe the safety instructions for “hold to run mode” (Section 2.4).

3.1.2.5 Interlocking

If two or more doors are interlocked, the control units are switched in that way that only one door can be opened at the same time. If one door is open or opening, all other doors are locked electrically and cannot be opened.

3.2 MCS buttons

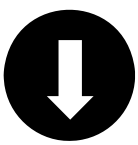
The three buttons “Open”, “Stop” and “Close” are designed to control the door and to reset the MCS in case of a malfunction. After clearing, the error message is reset automatically. A renewed Open or Close command starts the door movement. The five function keys below the display are used to activate the menu settings and change the values.



1) Display
2) Function keys
3) Open button
4) STOP button
5) Close button

3.2.1 Use of MCS buttons

The display shows messages, faults, etc. in plain text. Faults occurring during normal operation are indicated immediately (Section 6).

	**“Open” button** This button works as an “open” button, sending an opening pulse to the control system.
	**“Stop” button** The door can be stopped at any point using the “Stop” button. At the same time, the door is locked and can only be moved again once the “Open” or “Close” button on the control box or external actuators have been pressed. The stop function is shown on the display as plain text. The Stop button also initiates programming mode.
	**“Close” button** This button sends a closing pulse to the control system. There is a short delay before the closing sequence starts, because the safety systems are checked after every closing pulse.

3.2.2 Access code

The control system is protected by a freely selectable access code to prevent unauthorised access. Its activation is recommended to prevent unwanted changes of the parameters.

Details for setting and activation are described in Section 4.8.15.

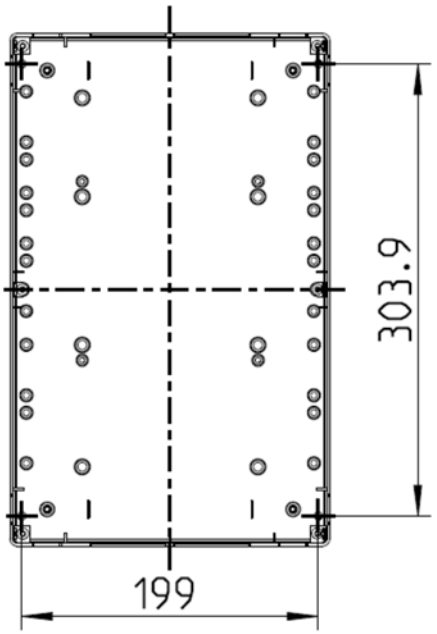
4 Installation

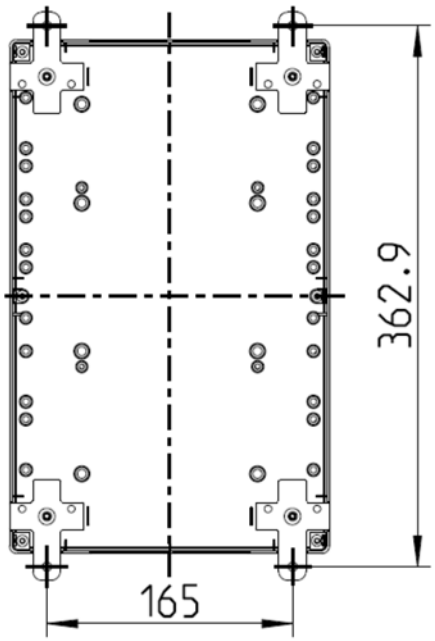
Check the general operation of the door and the safety devices for correct function after installation of the door.

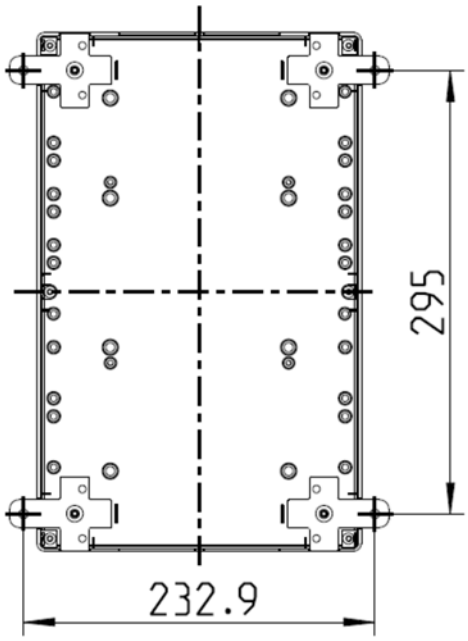
4.1 Mechanical installation of the control system/frequency converter

The control box should be mounted onto an oscillation- and vibration-free wall. Alternative attachment points outside the case are possible with the supplied mounting brackets. The frequency converter should be attached with the supplied mounting brackets near the drive unit directly onto the door or wall.

Mounting options of the control unit:







CAUTION

Risk of material damage

Improper installation can cause damage to the wiring.

⇨ Ensure that the wiring is routed correctly when installing the control box.

Unused entry points should be closed with appropriate sealed plugs. Only cable glands with at least IP 55 rating may be used. The upper cable inlets may be used only when no optional MCS units (see Section 8) are used, as otherwise the wires would block the assembly area for the slot-in modules.

4.2 Electrical installation of the control system

4.2.1 Electrical installation requirements

For correct operation of the control unit, the electrical installations must fulfil the following requirements:

CAUTION

It may be necessary to replace fuse F4 by the appended micro-fuse!

The wrong fuse could destroy the control unit

Use the right fuse for F4!

MCS-FC (230V) => F4 = 100mA medium slow

MCS-R (400V) => F4 = 80mA slow

- The bridge in the KL13 voltage selector must be inserted according to the control variant and mains voltage (see Section 4.2.5). A wrong insertion of the bridge can destroy the control system!
- All electrical work must be carried out according to the applicable local regulations and requirements.
- The following external pre-fuses must be provided on-site:

  MCS-R: 10-16A (B characteristic or comparable)

  MCS-FC: 10-16A (B characteristic or comparable)

  The pre-fuse must be selected according to the conductor cross-sectional area and the selectivity.

  Additionally observe the national statutory regulations!
- The door must be connected to the potential equalisation of the building.
- A general basic wiring diagram is included with each door. Wire the control system according to the requirements and set all parameters according to the specifications in the drawing for this type of door.
- If a residual current device (RCD) is used in combination with the MCS-FC (frequency converter), an AC/DC sensitive short-time delay RCD type B is mandatory. Only this type detects AC residual current, pulsating DC residual current as well as smooth DC residual current which could be caused by the

MCS-FC in the event of an error. Most European countries require RCDs with a rated residual current of max. 30 mA. The use of standard RCDs (type A) is not permissible.

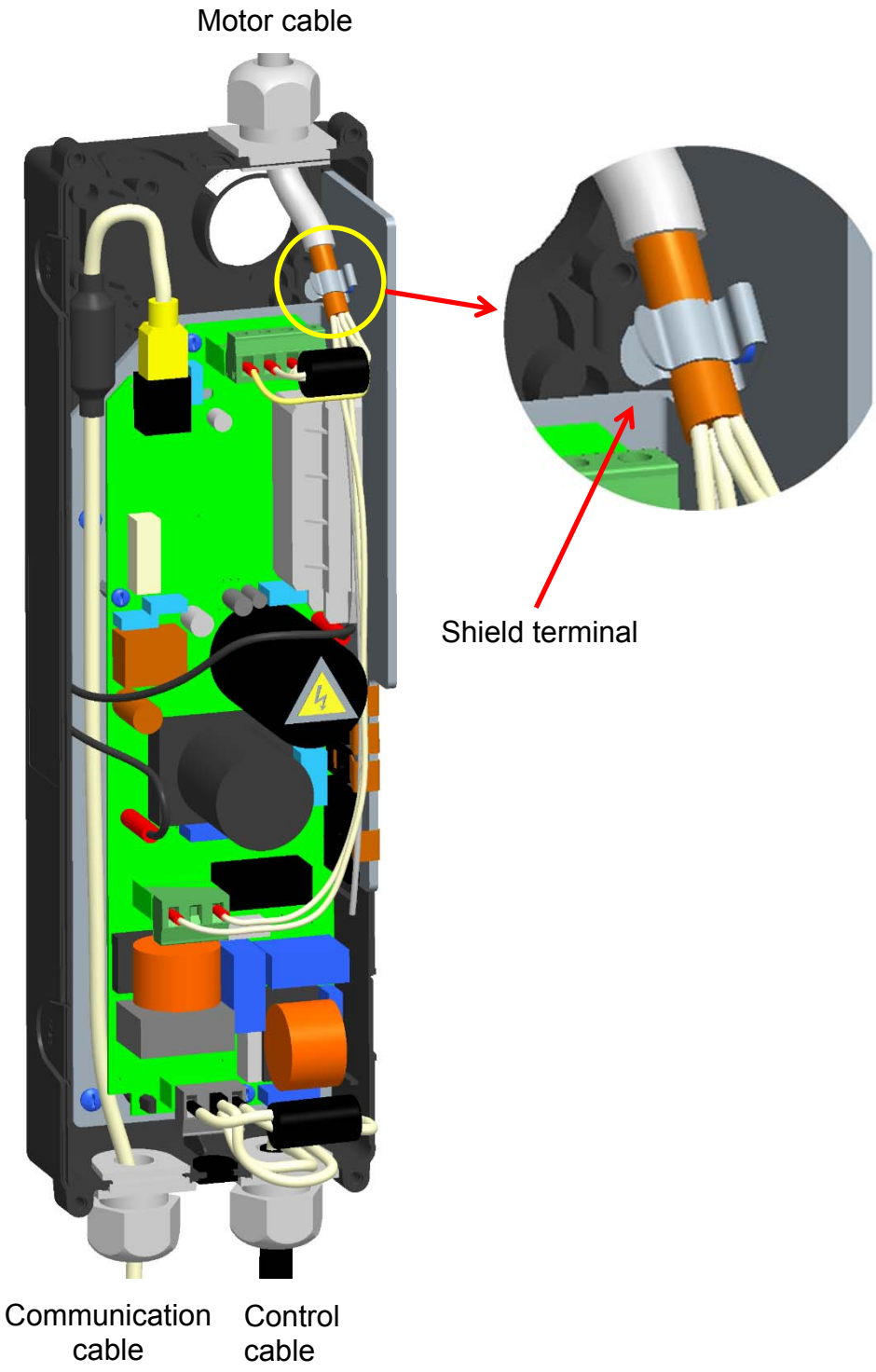
Additionally observe the national statutory regulations.
If you are in any doubt, please consult your electrician.

- The shield of the motor cable needs good and large-area contact in the frequency converter shield terminal. The white leads connected to the shield must be connected to the earth terminals. Otherwise the door could malfunction!
- The main supply cable must have a cross-sectional area of at least 1.5 mm².

4.2.2 Electrical connection

The electrical connect should be done according following default. The bare shield end of the motor cable should be clamped in the shield terminal of the frequency converter.

The cable inlet for the motor cable in the frequency converter needs to be break out.

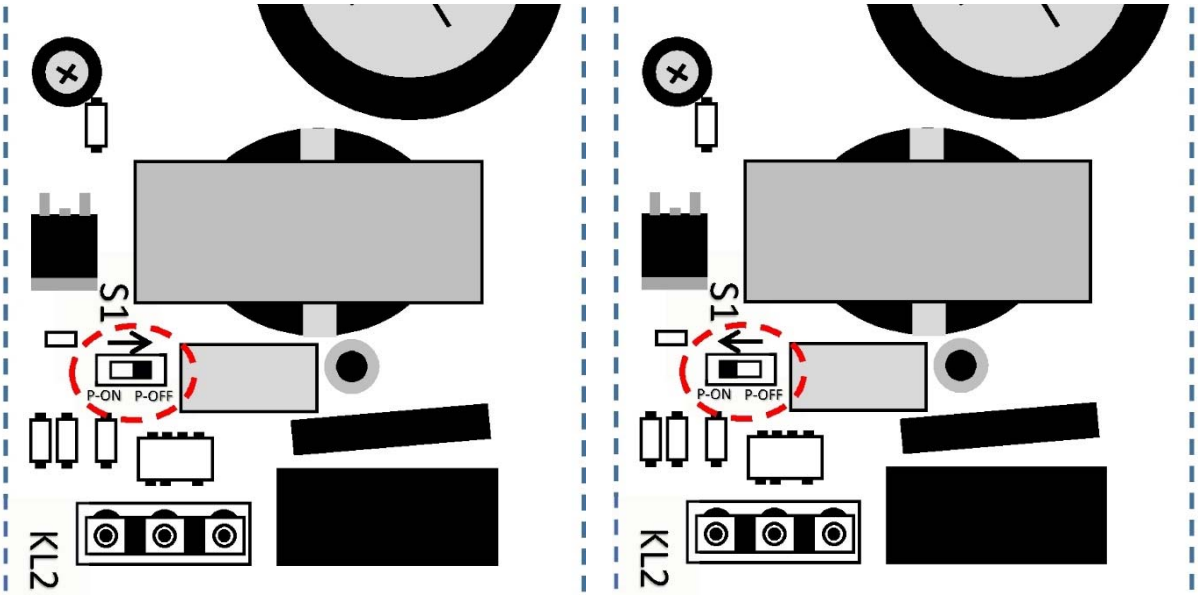


4.2.3 Selection of power-off or power-on brake (only for MCS-FC)

The operating mode of the installed brake can be set on the frequency converter board by means of a slider switch (S1).

A power-off brake is lifted when the operating voltage is applied. Without this voltage, its braking action is initiated.

A power-on brake initiates a braking action when the operating voltage is applied. It is lifted without an operating voltage.



Power-off brake setting

Power-on brake setting

4.2.4 Cross-section of the wiring / plug-in terminals

MCS-R:

Plug no.	Function	Without wire end sleeve	With wire end sleeve
KL1-8	I/O's	1.5 mm²	1 mm²
KL 9	24 V	1.0 mm²	0.34 mm²
KL10	Drive unit	2.5 mm²	2.5 mm²
KL11	PE	2.5 mm²	2.5 mm²
KL12	Supply voltage	2.5 mm²	2.5 mm²
KL13	Voltage selector	2.5 mm²	2.5 mm²
KL14 to 16	Output relay	2.5 mm²	2.5 mm²
KL17 to 23	I/O's	1.5 mm²	1 mm²

MCS-FC frequency converter

Plug no.	Function	Without wire end sleeve	With wire end sleeve
KL1	Drive unit	2.5 mm²	2.5 mm²
KL2	Brake	2.5 mm²	2.5 mm²
KL3	Supply voltage	2.5 mm²	2.5 mm²

4.2.5 Voltage selector

CAUTION

The voltage selector of the control unit (KL13) is incorrectly inserted.
Incorrect voltage selection on KL13 can destroy the control unit.

⇨ Ensure the correct setting of the bridge.

CAUTION

It may be necessary to replace fuse F4 by the appended micro-fuse!
The wrong fuse could destroy the control unit.

Use the right fuse for F4!

MCS-FC (230V) => F4 = 100mA medium slow

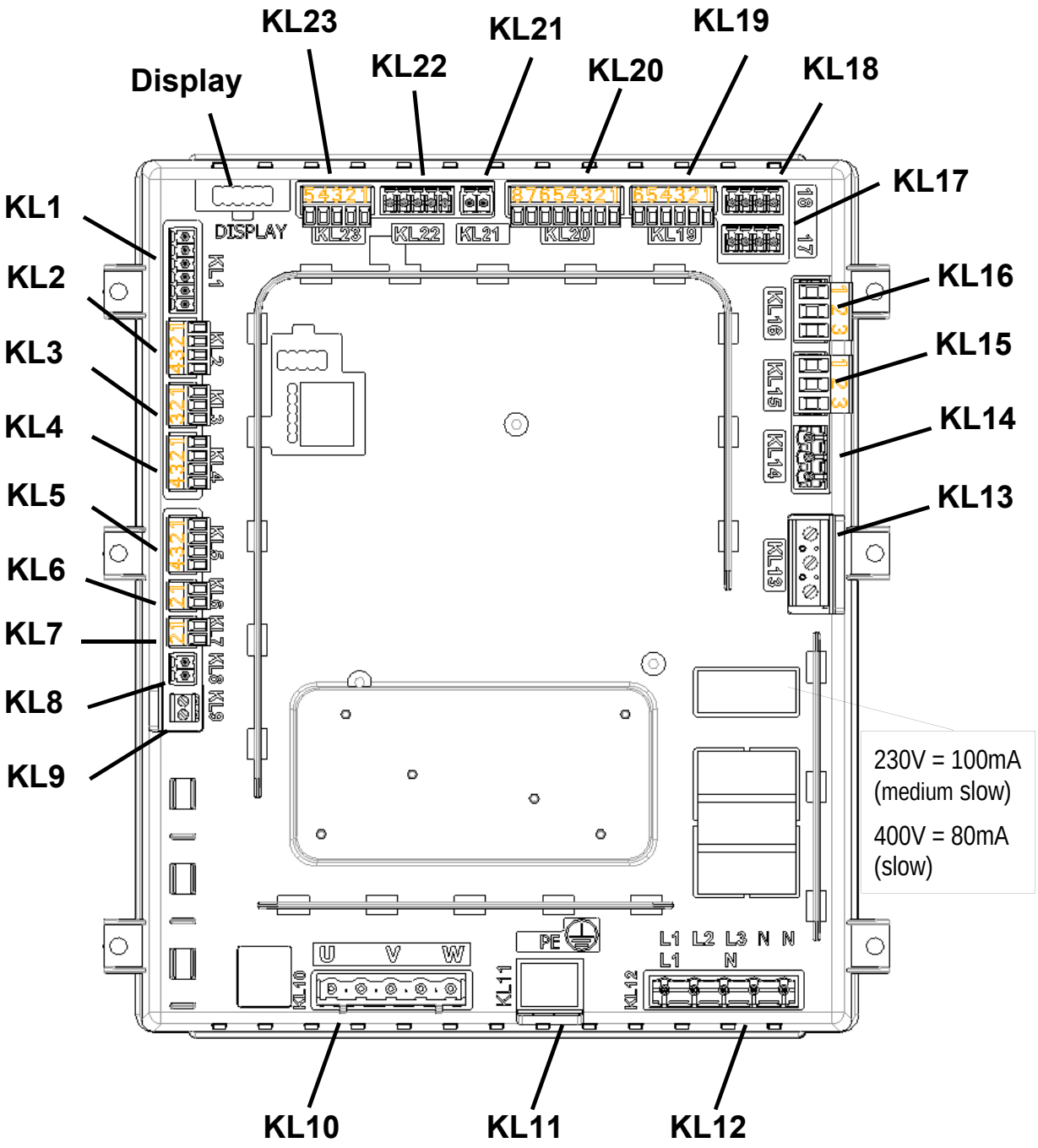
MCS-R (400V) => F4 = 80mA slow

The bridge in the KL13 voltage selector must be inserted according to the control version and main voltage.

MCS-R: (3x400V) => Bridge from 2 to 3

MCS-FC: (1x230V) => Bridge from 1 to 2

4.3 Terminal assignment of the control system



Terminal	Function	Pin assignment
KL1	Radio	1=GND, 2=24V, 3=Open/Stop/Close, 4=Open, 5=Close, 6=Stop
KL2	Safety edge (2x 8.2kΩ)	1/2=8.2kΩ, 3/4=8.2kΩ
KL3	Additional safety device (12V)	1=GND, 2=Signal, 3=+12V

KL4	Additional safety device tested (12V), pre-running safety photocell	1=GND, 2=Signal, 3=+12V, 4=Test
KL5	Door line photocell (24V), light curtain	1=+24V, 2=Test, 3=Signal, 4=GND
KL6	No function	
KL7	Additional stop	1=+24V, 2=Signal
KL8	Safety chain motor (thermal contact, emergency lever, emergency hand chain operator)	1=+24V, 2=Signal
KL9	24V output (max. 300mA)	1=GND, 2=+24V
KL10	Motor connection	MCS-R: 1=U, 3=V, 5=W MCS-FC: 3=V, 5=W
KL11	Earth connection (PE)	
KL12	Power supply	MCS-R: 1=L1, 2=L2, 3=L3, 4=N, 5=N MCS-FC: 1=L1, 3=N
KL13	Voltage selector	Bridge from 1/2=230V Bridge from 2/3=400V
KL14	Dry contact output relay (brake relay for MCS-R)	MCS-R: 1=COM and 3=NO for brake connection MCS-FC: 1=COM, 2=NC, 3=NO
KL15	Dry contact output relay	1=COM, 2=NC, 3=NO
KL16	Dry contact output relay	1=COM, 2=NC, 3=NO
KL17	Digital limit switch (motor)	1=RS485 Channel B, 2=RS485 Channel A, 3=GND, 4=+12V
KL18	Communication frequency converter	1=RS485 Channel B, 2=RS485 Channel A, 3=GND, 4=+12V
KL19	Input limit switch	1=upper end position, 3=lower end position, 5=red. opening height, 6=+24V
KL20	3-button station	1=Open, 2=+24V, 3=Stop, 4=+24V, 5= Close, 6= +24V, 7=Pulse (toggle function), 8=+24V
KL21	No function	
KL22	Connection for traffic light module	1=+24V, 2=GND, 3=SDA, 4=SCL, 5=+5V
KL23	Connection for loop detector module	1=24V, 2=GND, 3=COM, 4=Loop 1, 5=Loop 2

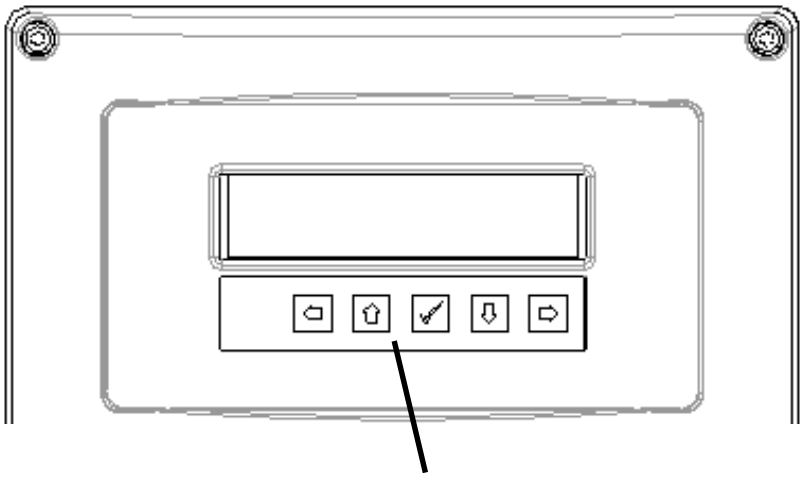
4.4 Digital limit switch

All Albany doors from ASSA ABLOY Entrance Systems with this control system type (MCS-R and MCS-FC) are delivered with a digital limit switch. It allows simple setting of the end positions and additional opening height directly in the control unit. Limit switches need not be set at the door.

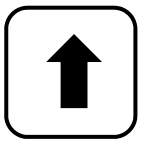
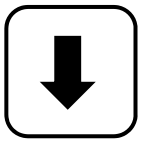
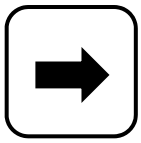
4.5 Emergency hand chain operator (KNH)

An optional emergency hand chain operator can be included in the safety chain at the digital limit switch in the motor. For this purpose, the bridge between clamps 21/22, 23/24 or 25/26 at the digital limit switch must be removed and the contact of the emergency hand chain operator must be connected here. This creates a series circuit with the thermo-contact and the optional emergency manual lever. The exact circuit may be seen in the basic wiring diagram.

4.6 Meaning of function keys



The five function keys below the display are active only in programming mode and have the following meanings:

 **SELECT**	**Function key 1 (←)** Menu selection (left)
UP  **BACK**	**Function key 2 (↑)** Change numbers and end position setting - 1x press, back to Menu selection - 2x press, back to Operating mode
NEXT  **CONFIRM**	**Function key 3 (√)** Confirm (save the input)
 **DOWN**	**Function key 4 (↓)** Change numbers and end position setting
 **SELECT**	**Function key 5 (→)** Menu selection (right)

4.7 Activate programming mode

Press the stop key for approximately 6 seconds to activate programming mode. Then enter the access code (default 00000) and confirm with the √ key. If no code is entered, entry can be skipped with the√ key.

4.8 Menu structure

CAUTION

Risk of personal injury and material damage

Set all listed parameters according the basic wiring diagram before using the door. A different setting of these parameters is not allowed and may be carried out only after consultation with our technical department!

If the door operates with the incorrect parameter settings, it may cause dangerous situations at the door.

⇨ Check all values for correct settings after programming.

The various menu items can be selected by pressing the ← or → key in the main menu. As an additional aid, the exact menu path to this item is shown in each submenu.

Menu item	Meaning	Setting value
LANGUAGE	Selecting and setting the menu language	See submenu 4.8.1
MOTOR SETUP	Selecting and setting all motor-relevant parameters	See submenu 4.8.2
ADJUST ENDPOS.	Setting and adjusting all door positions	See submenu 4.8.3
SAFETY DEVICES	Selecting and setting safety devices	See submenu 4.8.4
OPERATING MODE	Selecting and setting operating mode	See submenu 4.8.14
CODE ENTRY	Changing access code	See submenu 4.8.15
SPECIAL SETUP	Special setup (e.g. hold-open and close after passage times, reduced opening height, relay function, traffic light control, …)	See submenu 4.8.16
SERVICE	Setting and recall of all service-relevant parameters and data	See submenu 4.8.24

Press the √ key to select the selected menu. To exit programming mode, press the ↑ key.

4.8.1 Language

"English" is preset as the default language. In addition, you can select from the following languages:

Menu path:

LANGUAGE 

Menu item	Meaning	Setting value
DEUTSCH	German	---
NEDERLANDS	Dutch	---
ENGLISH	English	---
ITALIANO	Italian	---
FRANCAIS	French	---
ESPANOL	Spanish	---
SVENSKA	Swedish	---
POLSKI	Polish	---
ROMANA	Romanian	---
MAGYAR	Hungarian	---

The desired language can be selected with the ← and → key. Press the √ key to confirm and save. You will be returned automatically to the main menu.

4.8.2 Motor setup

CAUTION



Risk of personal injury and material damage

Set the end position and all listed parameters according the basic wiring diagram before using the door.

If the door operates with incorrect parameter settings, it may cause dangerous situations at the door.

⇨ Check all values for correct settings after programming.

The various parameters can be selected by pressing the ← or → key in the submenu.

Menu path:

LANGUAGE  MOTOR SETUP 

Menu item	Meaning	Setting value
MOTOR SETUP MOTOR DIRECTION	Setting and saving motor direction	See parameters listed in basic wiring diagram
MOTOR SETUP ENCODER DIR.	Setting and saving encoder direction	See parameters listed in basic wiring diagram
MOTOR SETUP MOTOR CONTROLLER	Selecting control version	See parameters listed in basic wiring diagram
MOTOR SETUP LIFT FORCE DET.	Selection for setting the power limitation	Do not use for Albany High Performance Doors from ASSA ABLOY Entrance Systems
MOTOR SETUP LIMITSWITCH TYPE	Selecting of digital limit switch version	See parameters listed in basic wiring diagram
MOTOR SETUP INV. PROFILE UP	Setting and saving converter parameters for opening movement	See parameters listed in basic wiring diagram
MOTOR SETUP INV. PRFL. DOWN	Setting and saving converter parameters for closing movement	See parameters listed in basic wiring diagram
MOTOR SETUP FU-PROFILE MOTOR	Setting the door drive	See parameters listed in basic wiring diagram

Select the parameter with the √ key and set it according to the parameters listed in the basic wiring diagram. Press the √ key to save. Querying is repeated for security reasons. Press the √ key to confirm or any other function key to cancel. After setting all values, press the ↑ key to leave the menu.

4.8.3 Setting end positions

CAUTION



Risk of material damage

The door could be moved manually beyond its upper and lower end position during installation (in “hold to run” mode). This may result in damage to the door.

⇨ Observe the door continuously when setting the end positions.

CAUTION

Risk of material damage

Before using the door, it is imperative to set the end positions and all listed parameters according to the setting values in the basic wiring diagram.

If the door operates with incorrect parameter settings, it may cause dangerous situations at the door.

⇨ Set all values according to listed parameters and check for correctness.

To make sure that the end position are not passed, the limits should only be changed in small steps!

When the door limits are being set, the door with MCS-FC will only creep in “hold to run” mode. The door with MCS-R will move during “hold to run” mode with normal speed.

The lowest permissible lower end position is reached when the mid-point of the rubber lip is at the level of the bottom plate of the side frames.

The various parameters can be selected by pressing the ← or → key in the submenu.

Menu path:

LANGUAGE 2X → ADJUST ENDPOS. ✓

Menu item	Meaning	Setting value
ADJUST ENDPOS. TOP	Setting and saving upper end position	---
ADJUST ENDPOS. BOTTOM	Setting and saving lower end position	---
ADJUST ENDPOS. BRAKE OFFSET	Start position for setting the brake point, with later automatic optimisation	See parameters listed in basic wiring diagram
ADJUST ENDPOS. FINE PITCH TOP	Fine adjustment of upper end position	Higher upper end pos => increase value with function key UP Lower upper end pos. => reduce value with function key DOWN

ADJUST ENDPOS. FINE PITCH BOTT.	Fine adjustment of lower end position	Higher bottom end pos. => increase value with function key UP Lower bottom end pos. => reduce value with function key DOWN
ADJUST ENDPOS. PRE END POSITION	Setting and saving of switch-off point for safety edge	See parameters listed in basic wiring diagram
ADJUST ENDPOS. SAFETY LIMIT	Setting and saving of switch-off point for safety limit switch	See parameters listed in basic wiring diagram
ADJUST ENDPOS. POS MED SPEED	Setting and saving of 2.5m switch point	---

Select the parameter with the √ key to set the end positions, move the door to the respective position and save the position with the √ key. The other parameter values can be set via the two function keys “UP” and “DOWN” and saved by means of the √ key. The current door position is displayed during setting as a numerical value for some parameters.

4.8.4 Setting of the safety devices

The various parameters can be selected by pressing the ← or → key in the submenu.

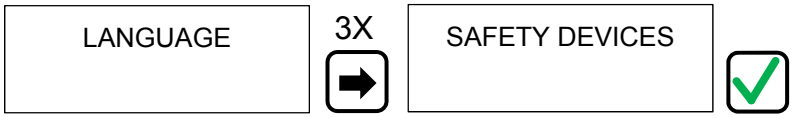


The current status is shown for the safety devices, e. g. DISABLED. This is marked as (State)* in the manual.



The closing movement in automatic mode always requires at least one closing edge safety device (contact edge or pre-running safety photocell) at KL2, KL3 or KL4. The exact settings for this purpose may be obtained from sections 4.8.8 and 4.8.10.

Menu path:

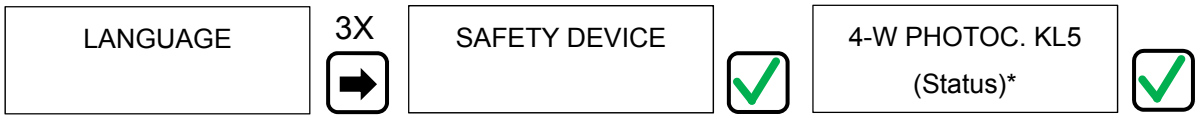


Menu item	Meaning	Setting value
4-W PHOTOC. KL5 (STATE)*	Selecting all relevant parameters for 4-wire photocell/light curtain KL5	See submenu 4.8.5
2-W PHOTOC. KL6 (STATE)*	Without any function	---
OSE1 SETUP KL3 (STATE)*	Selecting all relevant parameters for OSE1 KL3 (optical safety edge)	See submenu 4.8.8
OSE2 SETUP KL4 (STATE)*	Selecting all relevant parameters for OSE2 KL4 (optical safety edge and pre-running photocell (with test))	See submenu 4.8.9 – 4.8.13
SAFETY 1 KL2 1-2 (STATE)*	Selecting all relevant parameters for safety edge1 KL2 (safety edge)	See submenus 4.8.9 - 4.8.13
SAFETY 2 KL2 3-4 (STATE)*	Selecting all relevant parameters for safety edge2 KL2 (safety edge)	See submenus 4.8.9 - 4.8.13
REVERSE DELAY	Setting of reverse delay (only at MCS-R visible)	See parameters listed in basic wiring diagram
FINE PITCH DISABLE PHOTOC.	Fine adjustment of switch-off position for the 4-wire photocell. The value of the current switch-off pos. will be shown. Previous course setting as per 4.8.7	Higher switch-off pos. => increase value with function key UP Lower switch-off pos. => reduce value with function key DOWN

4.8.5 Selecting of 4-wire photocell / light curtain

The various parameters can be selected by pressing the ← or → key in the submenu.

Menu path:

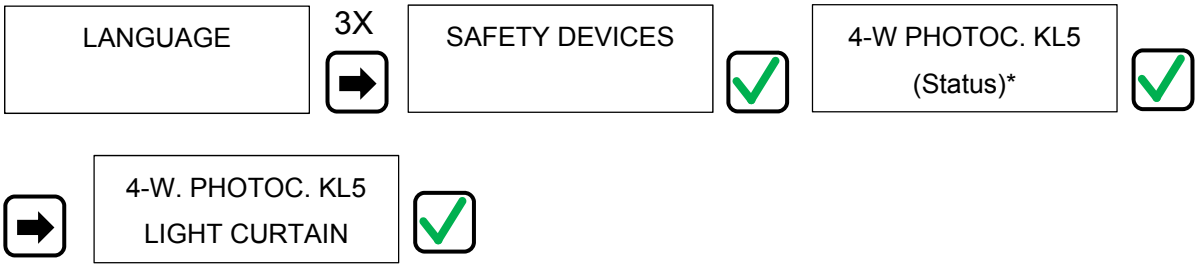


Menu item	Meaning	Setting value
4-W PHOTOC. KL5 DISABLED	4-wire photocell at KL5 disabled	---
4-W PHOTOC. KL5 LIGHT CURTAIN	Light curtain at KL5 activated	See submenu 4.8.6
4-W PHOTOC. KL5 UNTESTED	4-wire photocell at KL5 untested	See submenu 4.8.7
4-W PHOTOC. KL5 TESTED	4-wire photocell at KL5 tested (positive test pulse, test voltage on during test)	See submenu 4.8.7
4-W PHOTOC. KL5 TESTED INV.	4-wire photocell at KL5 tested (negative test pulse, no test voltage during test)	See submenu 4.8.7

4.8.6 Setting 4-wire light curtain

The various parameters can be selected by pressing the ← or → key in the submenu.

Menu path:

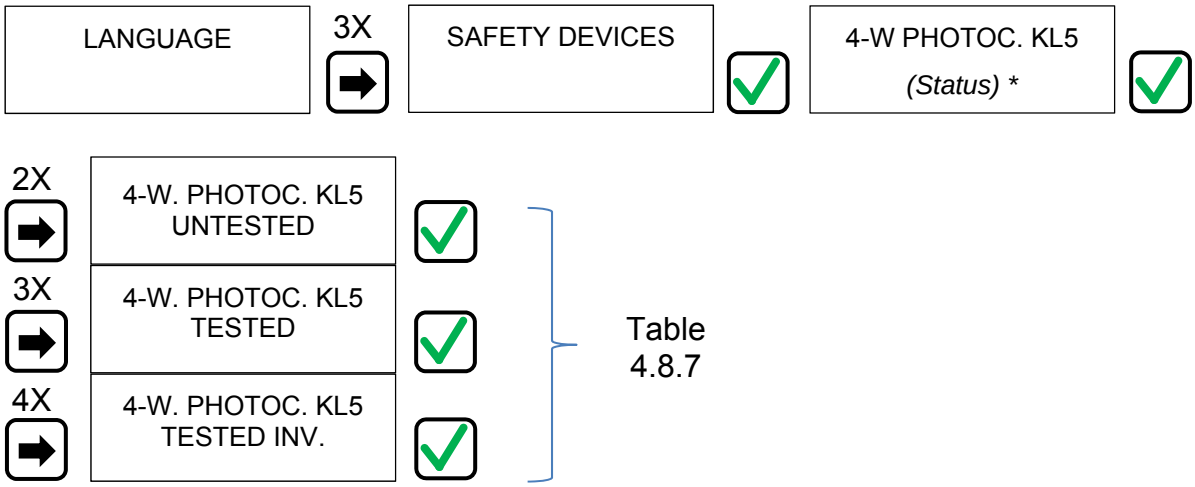


Menu item	Meaning	Setting value
4-W PHOTOC. KL5 DOWN FULL REV.	4-wire light curtain at KL5 with full reversal to the upper end position after activation	---
4-W PHOTOC. KL5 DOWN PART. REV.	4-wire light curtain at KL5 with partial reversal after activation	---
4-W PHOTOC. KL5 DISABLED	4-wire light curtain at KL5 disabled	---

4.8.7 Setting the 4-wire photocell

The various parameters can be selected by pressing the ← or → key in the submenu

Menu path:



Menu item	Meaning	Setting value
4-W PHOTOC. KL5 DOWN FULL REV.	4-wire photocell at KL5 with full reversal to upper end position after activation	See parameters listed in basic wiring diagram
4-W PHOTOC. KL5 DOWN PART. REV.	4-wire photocell at KL5 with partial reversal after activation	---
4-W PHOTOC. KL5 DOWN STOP	4-wire photocell at KL5 with door stop after activation	---
4-W PHOTOC. KL5 PULL-IN PROTECT	4-wire photocell at KL5 as pull-in protection with door stop	---
4-W PHOTOC. KL5 DOWN FULL/SW OFF	Bottom switch-off position of 4-wire photocell at KL5	See following note *1
4-W PHOTOC. KL5 PULL-IN MASKED	Top switch-off position of 4-wire photocell as pull-in protection	---
4-W PHOTOC. KL5 DISABLED	4-wire photocell at KL5 disabled	---

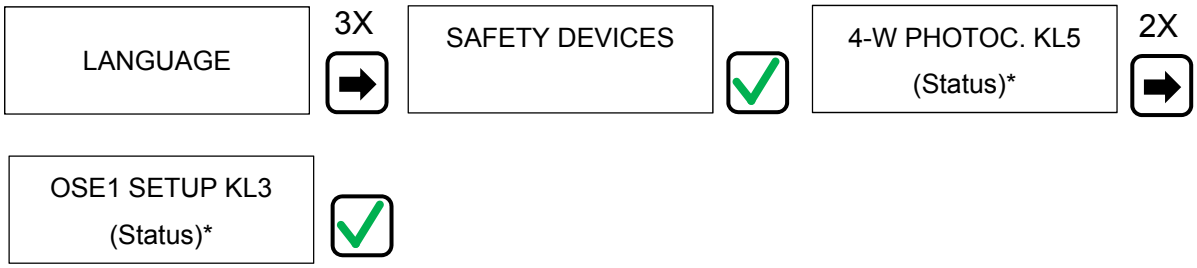
*1 To set the switch-off point of the doorline photocell, close the door and stop at the lower edge of the sealing lip at about 100-150mm (without coily cable), or 250-275mm (with coily cable) above the photocell. Save this value. If required, adjust it on the function control via the fine adjustment (see 4.8.4). For this purpose, increase the value via the function keys “UP” (higher switch-off point) or reduce it with “DOWN” (lower switch-off point).

4.8.8 Selecting the OSE/4 photocell terminal at OSE1

This submenu is available only when OSE 1 (KL3) is enabled. It is not displayed when this terminal is disabled.

This submenu allows the equipment connected to terminal KL3 to be selected by pressing the ← or → key. The pre-running photocell can be connected only at input KL4 (with testing)!

Menu path:



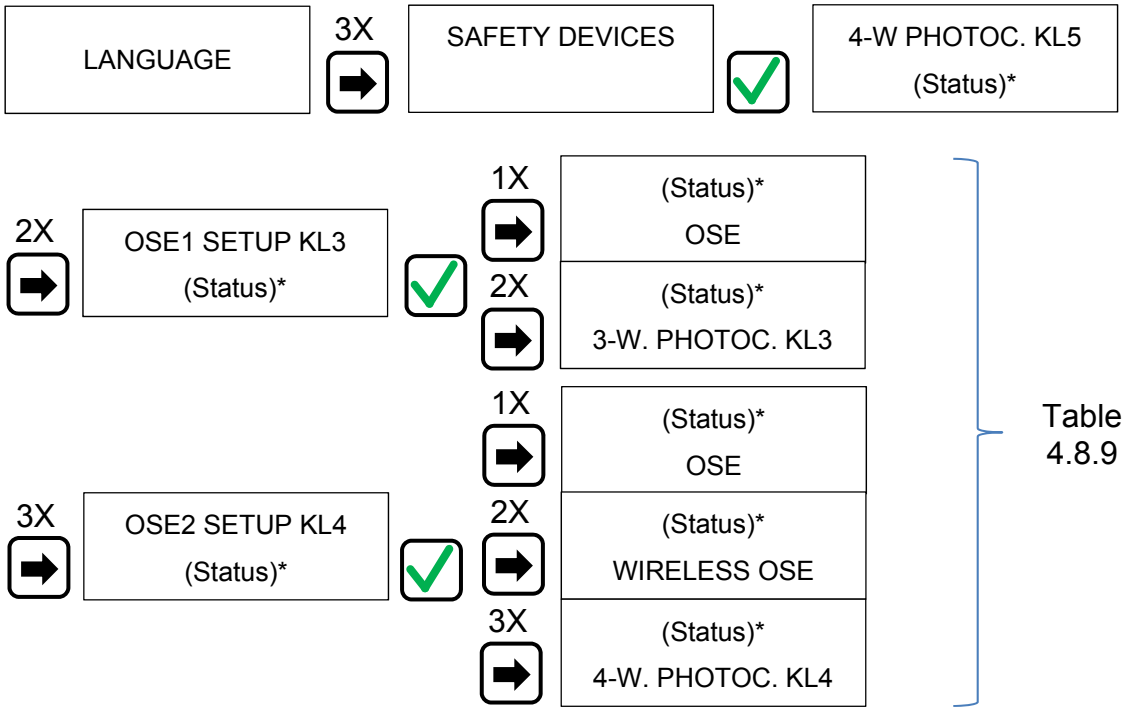
Menu item	Meaning	Setting value
(Status)* DISABLED	Disabling the OSE1 input.	---
(Status)* OSE	Selecting all relevant parameters for a safety device with OSE evaluation at the OSE1 input.	See submenu 4.8.9
(Status)* 3-W PHOTOC. KL3	Selecting all relevant parameters for a 3-wire photocell at the OSE1 input.	See submenu 4.8.9

4.8.9 Function of optical closing edge safety device (OSE1 and 2)

This submenu is available only with enabled OSE 1 (KL3) or OSE 2 (KL4). It is not shown if these are disabled.

By pressing the ← or → key in this submenu, the various parameters may be selected and the door function can be set with the activated safety device.

Menu path:



Menu item	Meaning	Setting value
(State)** DOWN FULL REV.	OSE with full reversal to upper end position after activation	See parameters listed in basic wiring diagram
(State)** DOWN PART. REV.	OSE with partial reversal after activation	---
(State)** DOWN STOP	OSE with door stop after activation	---
(State)** PULL-IN PROTECT	OSE as pull-in protection with door stop	
(State)** PULL-IN MASKED	Switch-off point for upper position of OSE as pull-in protection	---
(State)** DISABLED	Disabled OSE	---

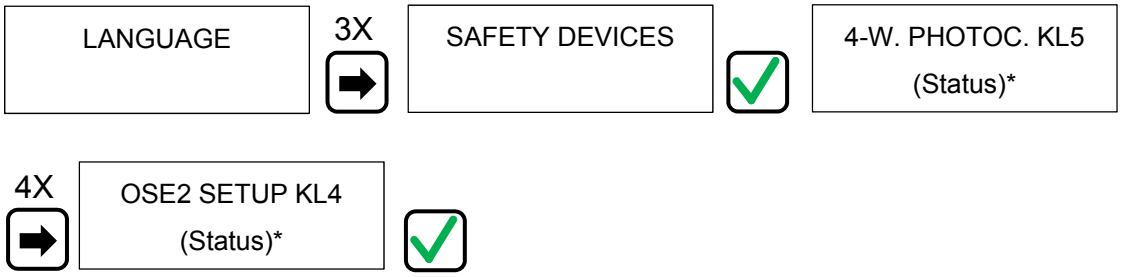
4.8.10 Selecting OSE/4-wire photocell terminal at OSE2

This submenu is available only with an enabled OSE 2 (KL4). It is not shown when OSE2 is disabled.

By pressing the ← or → key in the submenu, the various parameters can be selected and the door function set after activating the safety device.

The pre-running photocell can be connected only to the input KL4 (with testing)!

Menu path:



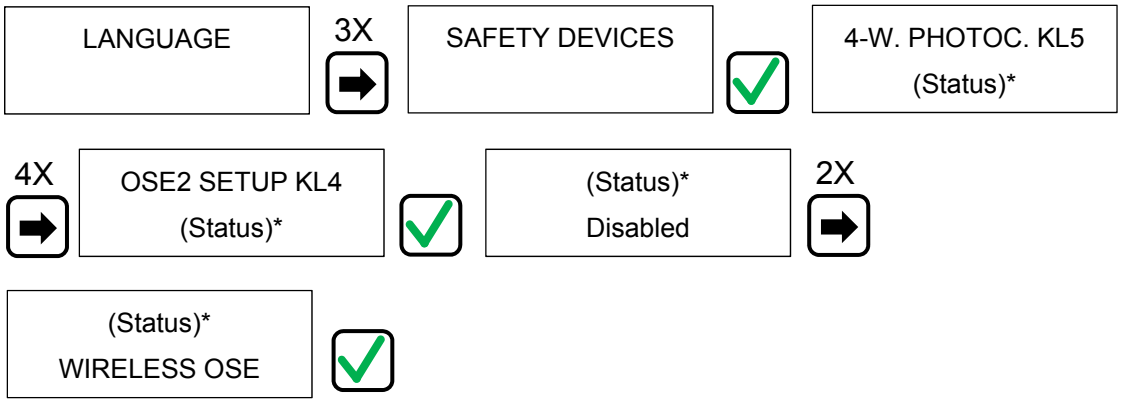
Menu item	Meaning	Setting value
(Status)* DISABLED	Disabling the OSE2 input.	---
(Status)* OSE	Selecting all relevant parameters for a safety device with OSE evaluation at the OSE2 input.	See submenu 4.8.9
(Status)* WIRELESS OSE	Selecting all relevant parameters for a safety device with wireless OSE evaluation at the OSE2 input.	See submenu 4.8.11
(Status)* 4-W. PHOTOC. KL4	Selecting all relevant parameters for a 4-wire photocell at the OSE2 input.	See submenu 4.8.9

4.8.11 Function of wireless OSE closing edge safety device at OSE2

This submenu is available only with enabled OSE 2 (KL4). When it is disabled, the submenu is not shown.

By pressing the ← or → key in this submenu, the various parameters can be selected and the door function can be set after activating a safety device.

Menu path:



Menu item	Meaning	Setting value
(Status)* DOWN FULL REV.	OSE closing edge safety device with full reversal up to the upper end position after activation	See parameters listed in basic wiring diagram
(Status)* DOWN PART. REV.	OSE closing edge safety device with partial reversal after activation	---
(Status)* DOWN STOP	OSE closing edge safety device with door stop after activation	---
(Status)* DISABLED	Disable OSE closing edge safety device	---

4.8.12 Setting safety contact edges 1 and 2

Connection safety edge 1 => KL2 1/2

Connection safety edge 2 => KL2 3/4

The various parameters can be selected by pressing the ← or → in the submenu.

Menu path:

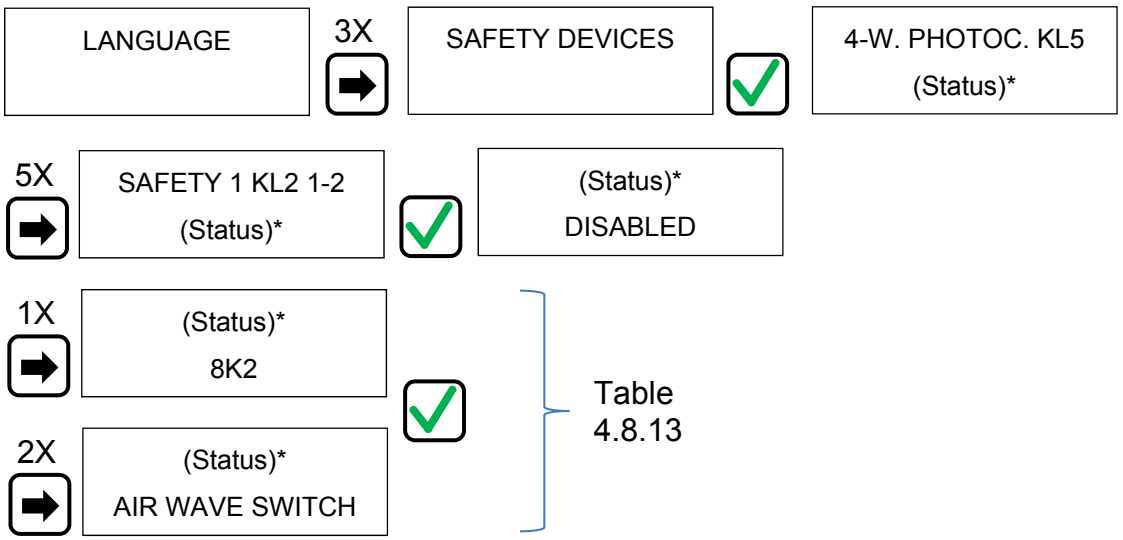


Menu item	Meaning	Setting value
(State)** DISABLED	Safety edge disabled	---
(State)** 8k2	Safety edge with 8.2 kΩ resistor	See submenu 4.8.13
(State)** AIR WAVE SWITCH	Safety edge with air wave switch	See submenu 4.8.13
(State)** 8K2 PASSDOOR	Safety edge with 8.2 kΩ for pass door	---
(State)** TESTED PASSDOOR	Radio safety edge for pass door	---

4.8.13 Setting safety contact edge 8.2 kΩ/air wave switch

The various parameters can be selected by pressing the ← or → key in the submenu.

Menu path:



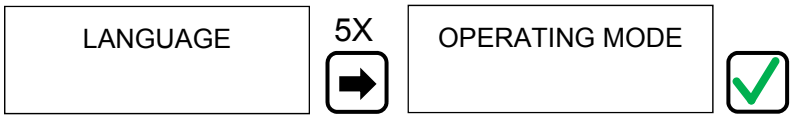
Menu item	Meaning	Setting value
(State)** DOWN FULL REV.	Safety edge with full reversal to upper end position after activation	See parameters listed in basic wiring diagram
(State)** DOWN PART. REV.	Safety edge with partial reversal after activation	---
(State)** DOWN STOP	Safety edge with door stop after activation	---
(State)** DISABLED	Disabled safety edge	---

4.8.14 Setting operation mode

The various parameters can be selected by pressing the ← or → key in the submenu.

Automatic operation mode is only possible with activated safety devices. Bridge the corresponding inputs in the control system if necessary (see basic wiring diagram) and activate the safety devices in the menu.

Menu path:



Menu item	Meaning	Setting value
OPERATING MODE AUTOMATIC ↑/↓	The door opens and closes in automatic mode	---
OPERATING MODE AUTO ↑/DEADMAN ↓	The door opens automatically and closes in “hold to run” mode	---
OPERATING MODE DEADMAN ↑/↓	The door opens and closes in “hold to run” mode	---

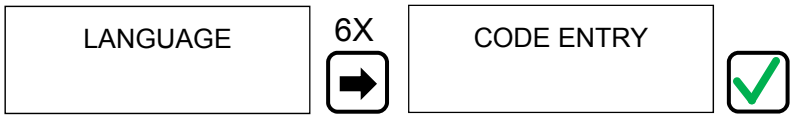
4.8.15 Setting access code

The access code can be changed in this submenu.



The factory setting is “00000”. If a new code is set, programming mode can only be accessed with the new code. Keep the new code in a safe place.

Menu path:

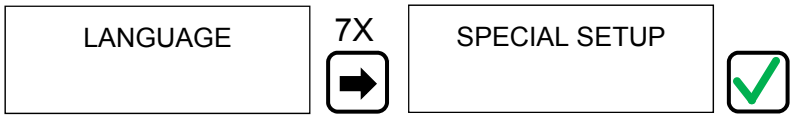


Press the ↑ and ↓ keys to change the individual values. Switch between the different digits with the ← and → keys. Saving is carried out once after full adjustment with the √ key. Saving is only possible on the last (right) code digit.

4.8.16 Special setup

The various parameters can be selected by pressing the ← or → key in the submenu.

Menu path:

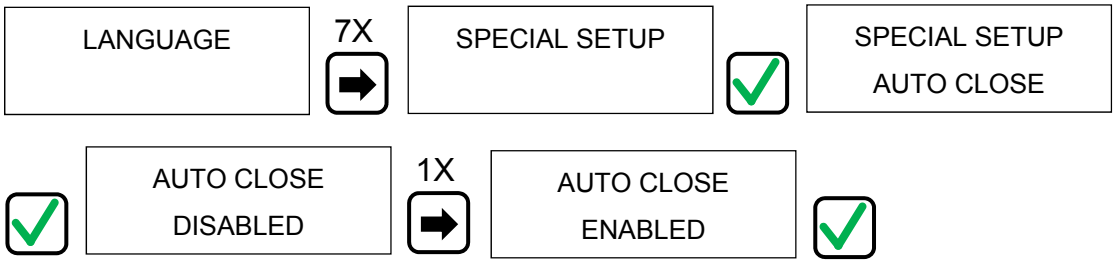


Menu item	Meaning	Setting value
SPECIAL SETUP AUTO CLOSE	Setting of hold-open and close after passage time as well as maximum activations of safety edge	See submenu 4.8.17
SPECIAL SETUP AUTO OPEN	Activation of automatic opening (useful e.g. for test run. An automatic open pulse is generated after a preset time if the door is closed. This is the same time as the hold open time. It cannot be set separately).	---

SPECIAL SETUP PARTIAL OPEN	Setting of reduced second opening height	See submenu 4.8.18
SPECIAL SETUP RELAYS	Selecting and setting of relay functions 1 to 3 (KL14, KL15, KL16)	See submenu 4.8.19
SPECIAL SETUP TRAFFIC LIGHT	Selecting and setting all relevant parameters for traffic lights	See submenu 4.8.23
SPECIAL SETUP CDM6	For internal use only	
SPECIAL SETUP DOOR CTRL MODE	Selecting door type (“sectional door” or “high-speed door")	See parameters listed in basic wiring diagram
SPECIAL SETUP EXTERNAL BUTTONS	Setting if external actuators are activated or deactivated in “hold to run” mode	See parameters listed in basic wiring diagram

4.8.17 Special setup automatic closing

Menu path:



The various parameters can be selected by pressing the ← or → key in the submenu. After activation of the “AUTO CLOSE” function, the following menu appears. Depending on the selection, not all submenus are visible.

CAUTION

Personal risk at the door is increased

Repeated collision with the door curtain can occur when increasing the value of the maximum number of safety edge activations

⇨ The value should only be altered by our service team in justified exceptional cases.

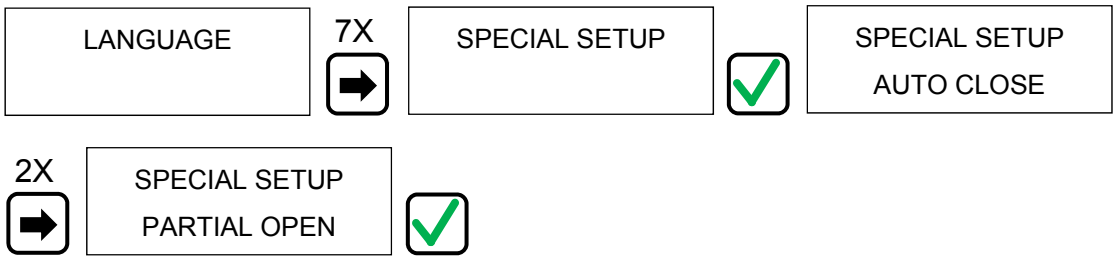
⇨ For safety reasons, the value of the electrical safety edge should only be increased to a maximum of 3 activations!

Menu item	Meaning	Setting value
CLOSE DELAY XX	Setting of hold open time	Setup range unlimited = no hold open time or 1-999 seconds
TRIG. AUTO CLOSE ENABLED	Deactivate or activate close-after-passage time (carwash)	---
CLEARING TIME 2	Setting close-after-passage time (carwash)	Setup range 0 to 999 seconds
SELECT TRIGGER BOTH	Setting which actuator starts the close-after-passage time (carwash CW). Radio, door line photocell or both are possible.	---
MAX EMERGN.-REV. 3	Max. activations of safety edge until the door is locked in its top position	See parameters listed in basic wiring diagram

4.8.18 Special setup for partial opening

A second opening height can be activated in this submenu.

Menu path:



The door starts its partial opening when an opening pulse is activated twice in succession. To set this parameter with the √ key, run the door to the corresponding partial opening position and save this position with the √ key. During setup, the current door position is shown as a numerical value. The open button on the control system or the input KL20;1 (3-button station, open command) can be used as an actuator for partial opening.

4.8.19 Special setup relay

The various functions of the three dry contact relays can be selected by pressing the ← or → key in the submenu. With a connected traffic-light module, the functions for relay 4 can be additionally selected (see also Section 8.2). This option is shown in the display only when the module is connected.

With MCS-R, relay 1 (KL14) is always used for brake control.

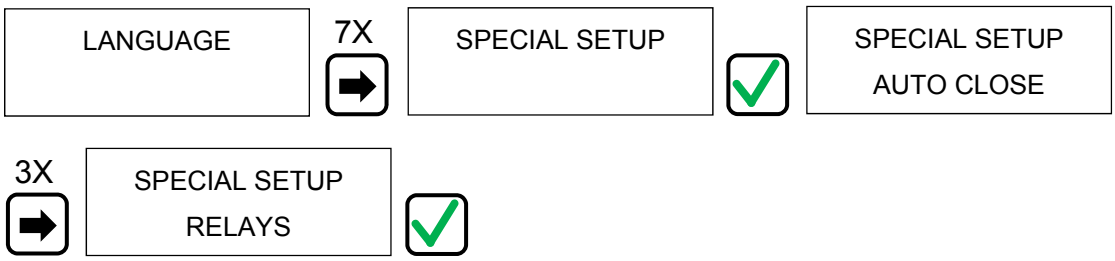
KL14=Relay 1

KL15=Relay 2

KL16=Relay 3

The menu structure and parameter setting is the same for all three relays. Only the setting for relay 1 is consequently shown.

Menu path:

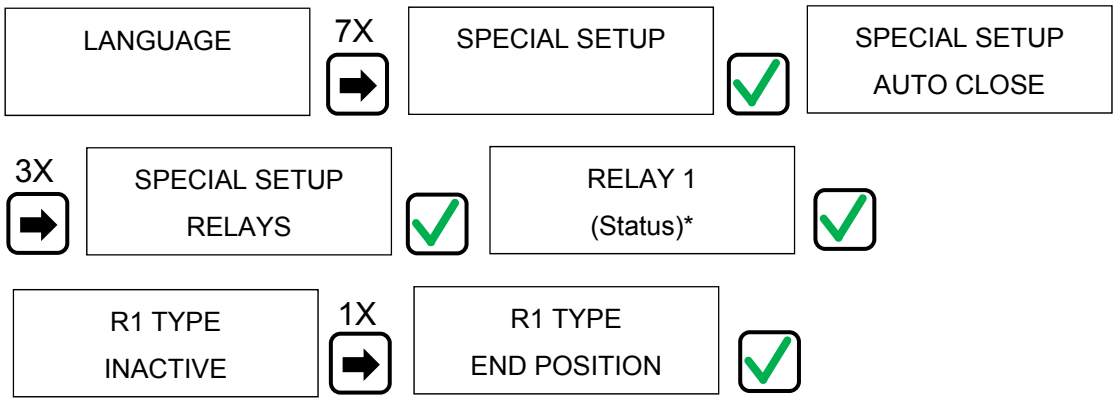


Menu item	Meaning	Setting value
R1 TYPE INACTIVE	Relay 1 not activated	---
R1 TYPE END POSITION	Relay 1 active in end positions	See submenu 0
R1 TYPE MOVEMENT	Relay 1 active during door movement	See submenu 4.8.21
R1 TYPE ELECTRICAL LOCK	Not used	---
R1 TYPE BRAKE	Relay 1 active for brake control (only MCS-R), setting of “RELEASE DELAY” according to parameters listed in basic wiring diagram	See parameters listed in basic wiring diagram
R1 TYPE RADIO	Not used	
R1 TYPE READY FOR WORK	Activation and setting for “READY FOR WORK” state	See submenu 4.8.22
R1 TYPE AUTO CLOSE	Setting of pre-running period before closing and function of relay (flash and permanent signal)	---

4.8.20 Relay 1 type end position

The various functions for the end positions of relay 1 can be selected by pressing the ← or → key in the submenu.

Menu path:

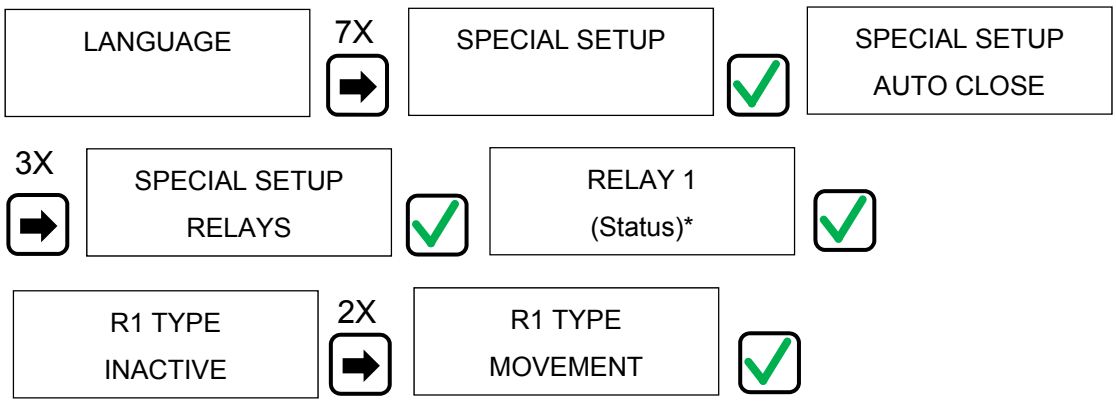


Menu item	Meaning	Setting value
R1 END POSITION ----	Switch point of relay for adjustable position. The pulse or permanent signal can be set in the submenu.	---
R1 END POSITION TOP	Switch point of relay for upper position. The pulse or permanent signal can be set in the submenu.	---
R1 END POSITION BOTTOM	Switch point of relay for lower position. The pulse or permanent signal can be set in the submenu.	---
R1 END POSITION BOTH	Switch point of relay for upper and lower position. The pulse or permanent signal can be set in the submenu.	---

4.8.21 Relay 1 type operation

The various functions used during operation of relay 1 can be selected by pressing the ← or → key in the submenu.

Menu path:



The first setting is the general pre-run time. The relay is inactive at setting 0s. Then select whether the activated relay flashes or is permanently activated.

Menu item	Meaning	Setting value
R1 DIRECTION ----	Switch point of relay for adjustable direction. Setting function of relay with door stop in the submenu is possible: pulse, permanent or switched off signal.	---
R1 DIRECTION UP	Switch point of relay during opening. Setting function of relay with door stop in the submenu is possible: pulse, permanent or switched off signal.	---
R1 DIRECTION DOWN	Switch point of relay during closing. Setting function of relay with door stop in the submenu is possible: pulse, permanent or switched off signal.	---
R1 DIRECTION BOTH	Switch point of relay during opening and closing. Setting function of relay with door stop in the submenu is possible: pulse, permanent or switched off signal.	---

4.8.22 Relay 1 type READY FOR WORK

The parameters for the “ready for work” state can be changed in this submenu. The message can be configured individually by setting the parameters accordingly. If more than one triggering criterion is to be selected, the number codes merely need to be added.

Example: Deactivation of “ready for work”:

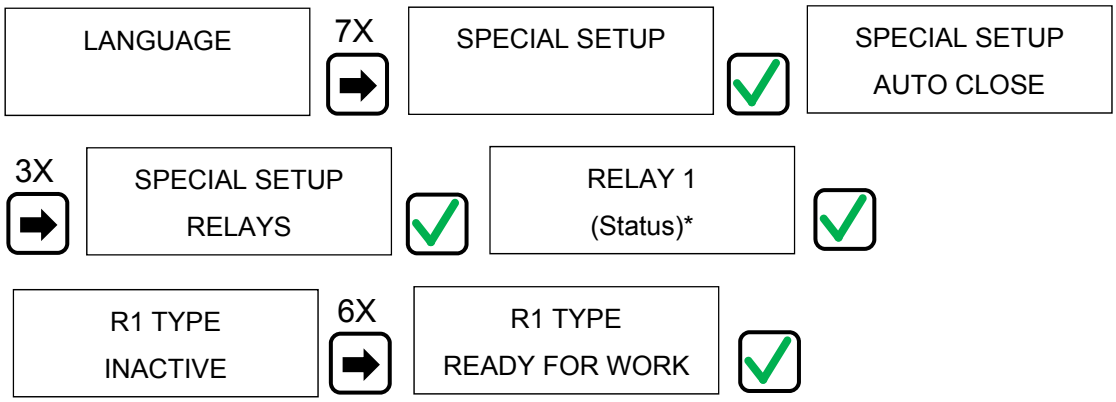
Activation of safety edge (safety edge 1) => 4

Emergency stop => 64

Safety chain motor => 128

Adding all three numbers = 196. This is set directly in the submenu.

Menu path:

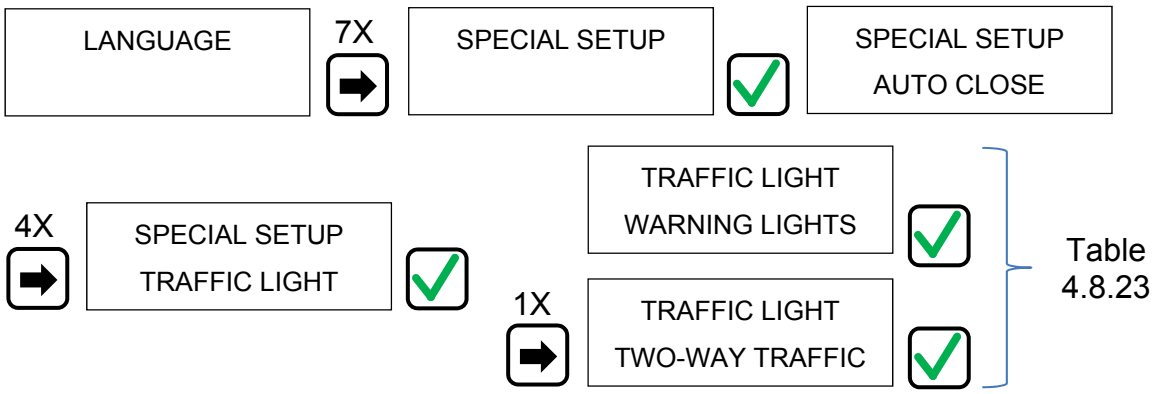


Trigger	Number code
Optical safety edge 1 (OSE 1 / KL3)	1
Optical safety edge 2 (OSE 2 / KL4) or pre-running photocell	2
Safety edge 1 (KL2 1/2)	4
Safety edge 2 (KL2 3/4)	8
4-wire photocell (KL5)	16
2-wire photocell (KL6)	32
Emergency stop (KL7)	64
Safety chain motor (KL8)	128
Pass door (KL2)	256
Sum of all triggers	511

4.8.23 Special setup traffic light control

The traffic light function with or without direction detection (two way traffic) can be selected by pressing the ← or → key in the submenu. The various times for the traffic light are set afterwards.

Menu path:

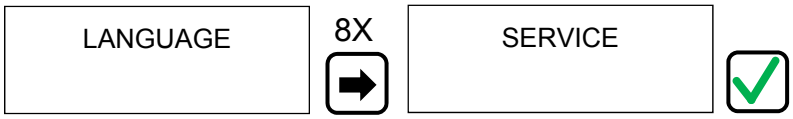


Menu item	Meaning	Setting value
TRAFFIC LIGHT LEAD TIME OPEN	Lead time before opening for the traffic light	---
TRAFFIC LIGHT OPENING TIME	Hold-open time of the door for traffic light control	---
TRAFFIC LIGHT LEAD TIME CLOSE	Lead time before closing for the traffic light	---
TRAFFIC LIGHT CLEARING TIME	Close after passage time of the door for traffic light control	---

4.8.24 Service settings

The various parameters can be selected by pressing the ← or → key in the submenu.

Menu path:



Menu item	Meaning	Setting value
SERVICE CYCLES	View of total cycles and days since installation. View of remaining cycles and days until the next service.	---
SERVICE SOFTWARE VERSION	View of software version	---
SERVICE CFG XML EXPORT	For internal use only	---
SERVICE SET TIME	Setting date and time	---
SERVICE MAINTE. COMPLETE	Reset service counter after service work	---
SERVICE MAINT. INTERVALS	Setting of service interval in cycles and days	See parameters listed in basic wiring diagram
SERVICE CLEAR ERRORS	Reset of error messages	---
SERVICE EVENT HISTORY	View event history	---

SERVICE CLEAR EVENT HIST	Clear event history	---
SERVICE CFG SOFT RESET	Resetting the configuration parameters to default settings. For a complete reset, the next menu item “SERVICE FACTORY RESET” must be performed additionally. After activation, control must be switched off briefly by pulling out the CEE plug so the settings are accepted. The event log will not be deleted.	---
SERVICE FACTORY RESET	Reset all parameters to factory settings. In addition, the previous menu item “SERVICE CFG SOFT RESET” should be activated so that the configuration parameters are reset. After activation, control must be switched off briefly by pulling out the CEE plug so the settings are accepted. The event log will not be deleted.	---

5 Operation

5.1 Safety instructions

Observe the safety instructions in Section 2 and in the user guide for the door.

5.2 Display

The display shows messages, faults, etc. in plain text. During operation, the operating mode and status information are shown.

5.3 Switch-on and reference run

The positioning of the door is regulated by a digital encoder mounted firmly on the drive unit. Switch on the control system.



In the frequency converter version, the parameterization of the converter is also shown briefly in the display.

1. After an open or close command, the door will run in standard operating mode.
2. The door is now ready for operation.

5.4 Opening the door

An “open” pulse is given by any actuator or from the keypad. The brake is released and:

MCS-FC: the motor is activated with a pre-defined acceleration ramp-up. The door opens with maximum speed until it reaches the starting point of the upper deceleration ramp. The door is now decelerated by the frequency converter and stopped in the adjusted upper end position. When this position is reached, the motor is switched off and the brake is activated. In doors with an operating current brake, the brake control takes place in reverse.

MCS-R: the motor is activated, stopped when the upper end position is reached and the brake is activated.

Depending on the pre-selected function, either the hold-open or close-after-passage time is started or the control system waits for a closing pulse.

5.5 Closing the door

Once the end of the hold-open or close-after-passage time is reached or a closing pulse is received, all connected safety devices are first tested by the control system. If this safety test is successful, the safety devices are activated, the brake is released and

MCS-FC: the motor is activated with a pre-defined acceleration ramp-up. The door closes with maximum speed until it reaches the starting point of the lower deceleration ramp. The door is now decelerated by the frequency converter and stopped in the lower end position. When this position is reached, the motor is switched off and the brake is activated. In doors with an operating current brake, the brake control takes place in reverse.

MCS-R: the motor is activated, stopped when the lower end position is reached and the brake is activated.

The safety devices are switched off max. 50 mm (approx. 1.9”) before the lower end position is reached. This prevents the door from reversing when it touches the ground.

5.6 Stopping the door

The door can be stopped in any position by pressing the “Stop” button. An “Open” or “Close” pulse then starts the door operation again. With activated stop, the hold-open time is not shown in the display.

5.7 Power failure

If a power failure occurs, doors with a power-off brake stop immediately wherever they are at that moment. Doors with a power-on brake move into the neutral position. This depends on the door version and the weight equalisation system. Once the main power returns, the door can return to normal operation directly after a renewed pulse.

5.8 Emergency manual operation

If there is a power failure, or repair or maintenance work is being carried out, it may be necessary to open the door manually.

CAUTION

The door may partially open or close by itself after the emergency manual lever has been activated!

There is a risk of injury by the door due to the movement of the door curtain.

⇨ Observe the door continuously when using the emergency manual lever to make sure that no persons or objects are present in the hazard area.

CAUTION

Risk of injury

There is a risk of injury due to simultaneous starting of the motor when the emergency hand crank is inserted.

⇨ Always switch the control system off before inserting the emergency hand crank.

Use of the emergency manual lever on the side frame can release the brake of the drive motor via a disengagement cable. Depending on the balancing system and the door position, the door curtain may then partially open or close by itself. It is then possible to reach below the bottom rubber strip in the middle of the door and push the door curtain up. As you do this, the door curtain will wind onto the top roll. Return the emergency manual lever to its original position. The control system can then be switched on and the door can be operated normally.

At doors equipped with an emergency hand crank, the door can be opened and closed by inserting it at the bottom of the drive unit and turning it.

The door must not be cranked via its upper or lower end positions.

If the emergency hand crank is still being operated when the main switch is switched on, the following message appears.

BLOCKED DOOR SAFETY

This message is cancelled automatically once the emergency hand crank has been returned to its original position or has been removed.



The door is not designed for regular manual use.

5.9 Service interval

If service is due, the following message is automatically shown:

PLANNED MAINTENANCE

The date when the next service is due can be called up on the display and reset after completion of maintenance (see Section 4.8.24).

5.10 Thermal switch

To protect the drive motor, three thermal switches are fitted in its windings. If the switches trip, the motor has become too hot and the drive is switched off immediately. The following message will then appear:

BLOCKED DOOR SAFETY

Once the motor has cooled down, the error is acknowledged automatically. A renewed Open or Close command starts the door movement. The door and the control system should then be checked for possible mechanical or electrical faults. The control system should not be switched back on until the motor has cooled down.

6 Fault correction

Contact the supplier or manufacturer for fault correction and repairs.

ASSA ABLOY Entrance Systems

Internet: www.assaabloyentrance.com

6.1 Event log readout

All error codes are recorded in the event log. A total of 128 events can be stored. When 128 events are reached, the oldest 64 events are deleted. The information is recorded in plain text with the date and time of the error.

6.2 Clear event history



The error log should only be cleared by ASSA ABLOY Entrance Systems Service, as all existing error messages are permanently erased.

See Section 4.8.24 for details.

6.3 View of software version

The used software version can be viewed in the display of the control system (see Section 4.8.24).

7 Technical data and control system label

7.1 General data

Housing material	Control unit: Plastic Frequency converter: Plastic
Dimensions of housing WxHxD without cable glands	Control unit: 215 x 345 x 140 mm Frequency converter: 106 x 355 x 106 mm
Mains voltage	MCS-R: 3/N/PE 400 V ± 10%, 50/60 Hz MCS-FC: 1/N/PE 230 V ±10%, 50/60 Hz
Max. motor power	MCS-R: 2.2 kW MCS-FC: 1.1 kW
Operating temperature range	-15°C - +50°C
Protection class	IP 54
Internal fuses	10A slow main power fuse (F1 – F3) 100mA medium slow fuse (F4) for 230V power supply. 80mA slow (F4) for 400V power supply
External fuse supplied by customer	MCS-R: 10-16A (B characteristic or comparable) MCS-FC: 10-16A (B characteristic or comparable) (1.1kW version)
Residual current protection (RCD) (only if needed)	MCS-FC: AC/DC sensitive short-time delay type B Observe the instructions in Section 4.2.1!
Residual current caused by MCS	≤ 1mA according to EN 60335-1
Control voltage / external supply	12/24 VDC / max. 300mA
Brake voltage	MCS-R: 96 / 196V DC (selection with used rectifier) MCS-FC: 96/196V DC (selectable via terminal KL2 in the frequency converter and power-off and power-on brake (selectable via sliding switch S1 in frequency converter). Max. 300mA
Dry relay contacts	MCS-R: 3 items on control board, function selectable min. 10mA, max. 230V AC / 4A (1 item used for brake control) MCS-FC: 3 items on control board, function selectable min. 10mA, max. 230V AC / 4A
Weight	MCS-R: approx. 2.2 kg MCS-FC: approx. 1.3 kg (only frequency converter)

7.2 Labelling

The nameplate of the control unit is located on the right side of the control system housing.

Spanning 230V/400V Voltage 3N~ 50 Hz	Typ MCS-R1 Type 8930R0245	Driftart 100% Duty type	Kapslingsklass IP 54 Degree of protection
Strömstyrka 0,8A Current	Tillverkningsar 2016 Year of manufacture	Sene nr/Serial no 1160215890	Vndmoment Torque
Ineffekt Input power	Driftstid Operating time	Driftcykler Operating cycles	Manoverspanning 24V DC Control voltage
Temperaturomrade Temperature range	ASSA ABLOY Entrance Systems AB Lodjursgatan 10 S-261 44 Landskrona Sweden		

The nameplate of the frequency converter is located on the right side of the frequency converter housing.



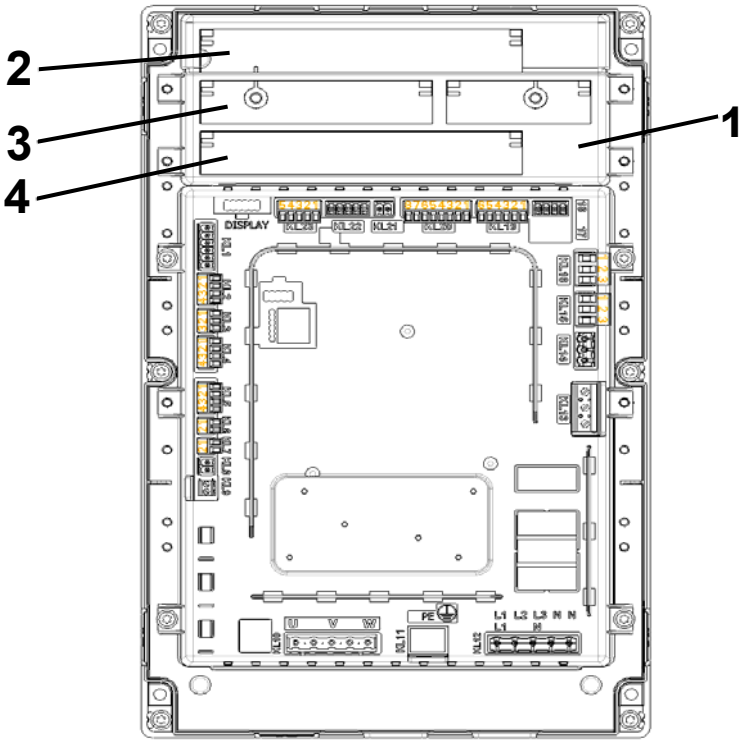
8 Optional MCS units

The MCS control system enables various plug-in modules to be directly installed. These are inserted into the designated plug-in points and connected to the MCS with the enclosed adapter cable.

8.1 Module bracket

The module bracket is installed directly above the main board in the control housing. It is inserted into the designated plug-in point and connected to the MCS with the enclosed adapter cable.

The upper cable inlets may be used only when no optional MCS units are used, as they would otherwise block the insertion space of the slot-in module.



1.) Module bracket
2.) Slot for traffic light module
3.) Slot for loop detector module
4.) Slot for radio receiver module

8.2 Traffic light module

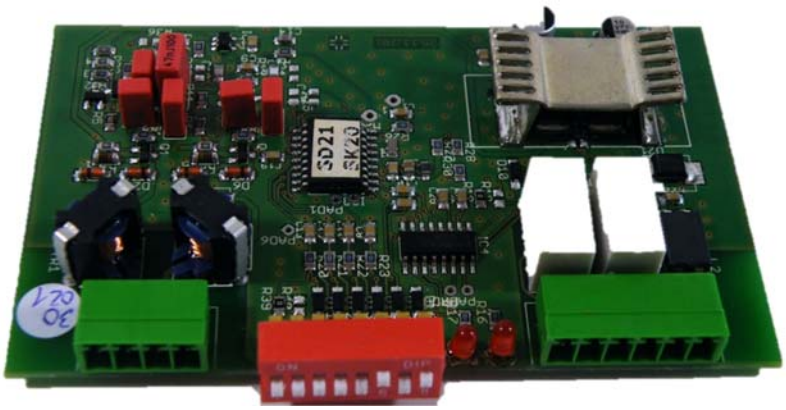


The traffic light module has four integrated relays. The connection is at terminal KL22 in the control box. Detailed adjustments are described in the supplied manual and the wiring is shown in the basic wiring diagram.

Terminal	Function
X3-1	Connection power supply traffic lights (disconnect with relay K2)
X3-2	Red traffic light, inside
X3-3	Green traffic light, inside
X3-4	Red traffic light, outside
X3-5	Green traffic light, outside
X3-6/7/8	Connection power supply for traffic lights
X3-9	Dry contact relay K4 (NC)
X3-10	Dry contact relay K4 (NO)
X3-11	Dry contact relay K4 (COM)

Terminal	Function
X4-1	GND
X4-2	+24V
X4-3	+5V
X4-4	SCL
X4-5	SD

8.3 Loop detector module



This is a double detector to connect two loops. The connection to the MCS is done at terminal KL23 in the control box. Detailed adjustments are described in the supplied manual and the wiring is shown in the basic wiring diagram.

Terminal	Function
1/2	Loop 1
3/4	Loop 2

Terminal	Function
1/2	Switching output loop 1
3/4	Switching output loop 2
5	+ 24 V DC
6	GND

Setting DIP switches:

DIP switch	Function
1	Frequency adjustment
2	Frequency adjustment
3	Sensitivity loop 1
4	Sensitivity loop 1
5	Sensitivity loop 2
6	Sensitivity loop 2
7	Direction detection
8	Frequency boost (truck with build-up)

8.4 Radio receiver module

This is a four-channel radio receiver. The connection is done at terminal KL1 in the control box. Detailed adjustments are described in the supplied manual and the wiring is shown in the basic wiring diagram.

CAUTION

A close command from a hand-held transmitter can lead to a dangerous situation if it is made without a clear view of the door.

When several actuators are used, the door operator should be aware of possible unexpected door movements and possible collisions with the door.

⇨ A close command should not be used with a radio actuator.

Brief guide

To programme the transmitter, jumper JR5 on the receiver card should be on OFF (jumper connection open), as otherwise all four transmitter channels will be simultaneously saved at the transmitter. When this is on OFF, only the pressed channel key will be saved on the receiver. The four jumpers JR1, JR2, JR3 and JR4 connect the four channels with the output relay.

Recommended configuration (open by radio pulse, close after setting the control function, e.g. close after a set time):

JR5=OFF

JR6=OFF

JR1=OUT1, JR2=OUT2, JR3=OUT3, JR4=OUT4

DIP switches 1-4=OFF

Programming the transmitter in the receiver:

1.) Press the PRG key on the receiver briefly until LED SIG lights up.

2.) Press the desired key on the transmitter for 2 seconds within the receive range. During adaptation in response to an activated transmitter, LED SIG flashes slowly. After adaptation, LED SIG flashes rapidly as confirmation. The transmitter is saved. If the LED lights up again, another transmitter can be programmed. The transmitter must be activated within ten seconds, as the unit otherwise automatically exits programming mode again (LED SIG goes off).

3.) Programming stops automatically ten seconds after the last save or when the PRG key is pressed again.

4.) To programme additional transmitters, press the PRG key again briefly (LED SIG lights up) and continue as described in point 2 above).

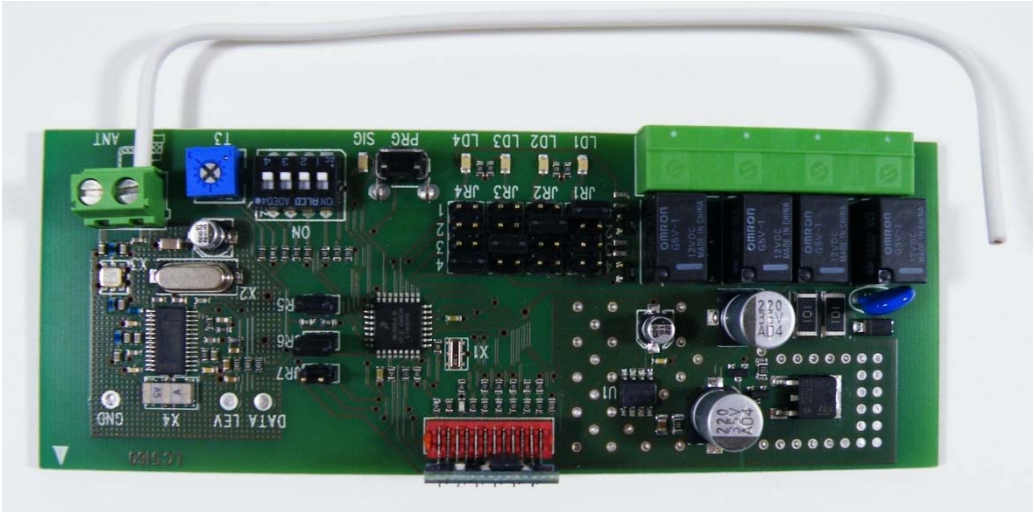
Deleting transmitters from the receiver:

1.) Hold the PRG key on the receiver for 3 seconds until LED SIG flashes slowly.

2.) To delete all transmitters, press the key for another 3 seconds.

3.) To delete individual transmitters, press a key of the transmitter to be deleted. Individual channels cannot be deleted.

4.) Deleting is confirmed by rapid flashing of LED SIG.



Channel 1 (jumper JR1)	Pulse (Open/Stop/Close)
Channel 2 (jumper JR2)	Open
Channel 3 (jumper JR3)	Close
Channel 4 (jumper JR4)	Stop

Hand-held transmitter:



8.5 Additional power supply

The control system has an internal 12 / 24 V DC power source which can be loaded with max 300mA. The 24 V control voltage can be extended with 630 mA by an additional power supply

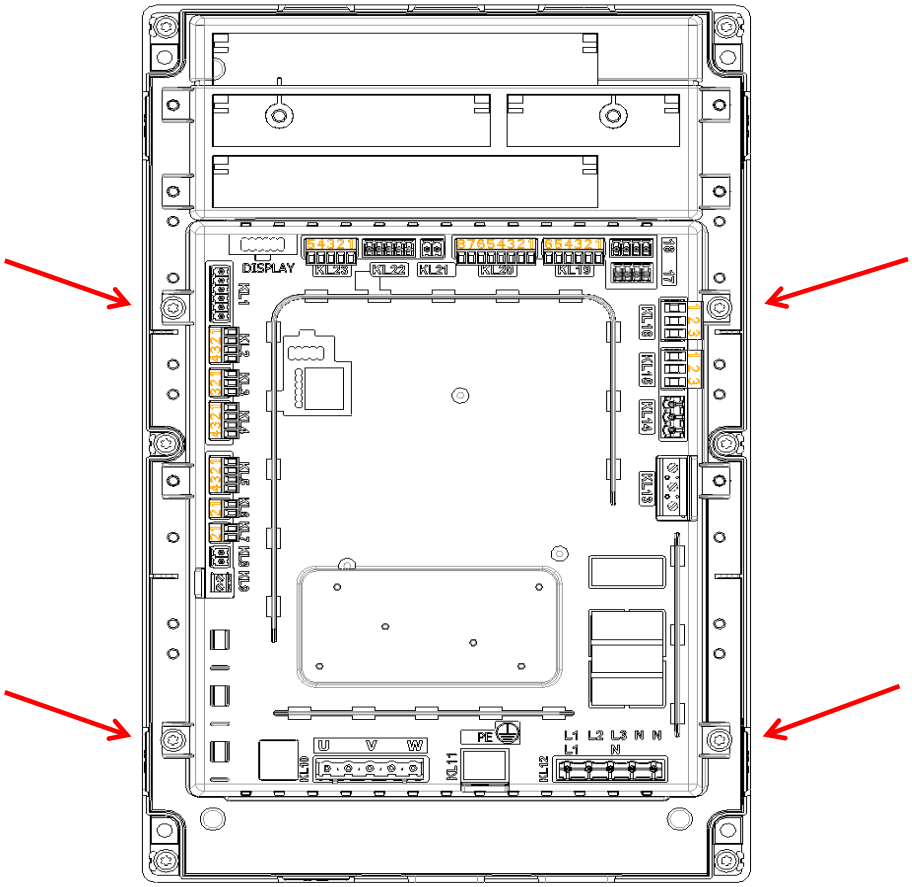
WARNING

Risk of electric shock

Unprotected, live components are located beneath the rack.

⇨ Before performing any work on electrical appliances they must be switched off.

The plug-in power supply is mounted on the DIN rail below the MCS rack. For installation of the power supply the rack has to be dismantled. Remove the 4 Torx-screws (TX20) (see figure below) and pull-out the rack.



There is an integrated DIN rail on the backside of the rack. The power supply can be snapped on this (see figure below).

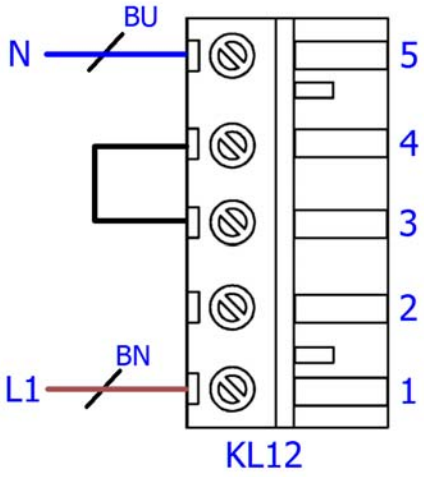
The two power supply plugs for the electrical connection can be plugged on the PCB-board. Due to their design they are protected against reverse polarity and mixed up.



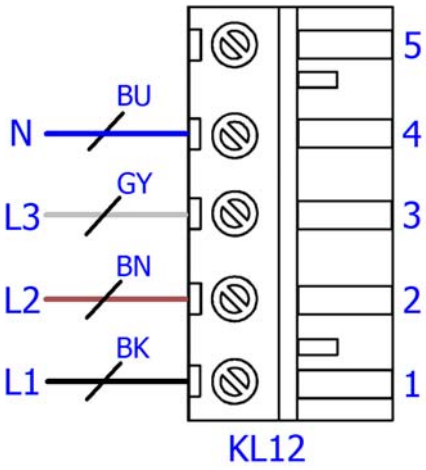
Power supply mounted on DIN rail

After the power supply is mounted and the two connectors are plugged, the rack should be mounted and fixed again.

To connect the additional power supply at the frequency converter version (230V) to the net voltage the following connection of the net cable on KL12 is necessary. The connection between pole 3 and 4 is important! At the relay version (400V) the connection on KL12 should not be changed.



Frequency converter version 1x230V



Relay version 3x400V